Original summary:
The story begins in thick jungle on Sekk, which we are told is a "second moon" which retains a "breathable atmosphere" around a lake surrounded by eleven jungled valleys. In this way, it is implied that Sekk is a second moon of Earth. In the jungle, we meet Noork and a young woman named Sarna. They begin traveling together through the jungle, but soon Sarna disappears and Noork is attacked. This is our first encounter with the Misty Ones, who blend in with the jungle foliage. Noork defeats the Misty Ones and continues toward the lake and island where they make their home. Noork briefly encounters his friend Ud near the marshy lowlands that lie between the jungled valleys on Sekk and the central Lake of Uzdon, but this area is not described. When Noork reaches the central island in the lake, we encounter a non-jungle landscape for the first time. Noork finds himself in a cultivated field, and sees the shape of a huge white skull about half a mile away. After speaking with an enslaved man and learning where Sarna is being held, Noork continues toward the skull. The skull is a dome of white stone, with black stone for eye-sockets and nose-holes. The interior contains a raised altar made of precious metals–gold, silver, and brass–and precious stones, as well as stone images of the two gods the Misty Ones worship. Below the altar is the caged area where the young women are held; Noork detects the entrance to this area by its foul odor. The room where the young women are kept is dimly lit by only two torches, very damp with pools of dirty water all around, and holds at least twenty young women. They have nothing to sit on but rotten grass mats. In contrast to the enslaved men who are out in the cultivated fields and open air, the young women are in a desperate situation indeed. They can only sit in their foul, rotting prison and wait to be sacrificed.

Perturbed summary (Summarized):
The story begins on Sekk, a second moon with a breathable atmosphere around a lake and eleven jungled valleys. Noork meets Sarna in the jungle, but she soon disappears, and he is attacked by the Misty Ones, whom he defeats. Noork encounters his friend Ud near marshy lowlands before reaching the central island, where he finds a cultivated field and a large white skull. After speaking with an enslaved man about Sarna's location, Noork approaches the skull, which is a dome of white stone with a raised altar inside, containing precious metals and images of the Misty Ones' gods. Below the altar, he discovers a dimly lit, damp room holding at least twenty young women, who are imprisoned and awaiting sacrifice.

Figure 5: Example of an original summary from SQuALITY and its Summarized perturbation.

Original summary:
The Court held that there had been a violation of Article 9 of the Convention, finding that the applicant did not enjoy the necessary procedural safeguards in having his request for alternative civilian service examined. The Court considered in particular that the Greek authorities had failed in their duty to ensure that the interviewing of conscientious objectors by the army's Special Board took place in conditions that guaranteed procedural efficiency and the equal representation required by domestic law. In this respect, it noted that: the applicant had been interviewed by a Board made up primarily of servicemen, two of the civilian members of the Board being absent but not replaced; the Minister of Defence's final decision, on the basis of a draft ministerial decision following the Board's proposal, did not afford the requisite safeguards of impartiality and independence; the scrutiny of the Supreme Administrative Court concerned only the lawfulness of the decision, not the merits, and was based on the assessments of the Special Board.

Perturbed summary (Simplified):
The Court found that Article 9 of the Convention was violated. The applicant did not have the necessary procedural safeguards for his request for alternative civilian service. The Court noted that the Greek authorities did not ensure that the army's Special Board interviewed conscientious objectors in a way that met procedural efficiency and equal representation as required by law. The applicant was interviewed by a Board mostly made up of servicemen, and two civilian members were absent without replacement. The Minister of Defence's final decision was based on a draft from the Board and did not ensure impartiality and independence. The Supreme Administrative Court only reviewed the lawfulness of the decision, not its merits, and relied on the Special Board's assessments.

Figure 6: Example of an original summary from LexAbSumm and its Simplified perturbation.

Original summary:
Solving large complex partial differential equations (PDEs), such as those that arise in computational fluid dynamics (CFD), is a computationally expensive process. This has motivate the use of deep learning approaches to approximate the PDE solutions Convolutional neural networks (CNNs) have been shown to achieve a speedup of between two and four orders of magnitude over numerical flow solvers [40-42]. Data-driven neural solvers learn from observations of fluid dynamics, usually from large datasets generated by numerical simulation, without (in general) any knowledge of the governing equations. The type of DL model employed for data-driven solvers primarily depends on the nature of the physical problem, with CNNs being a natural choice for simulations where the geometry aligns with the coordinate system, while GNNs are preferred for simulations of more complex geometries and Lagrangian systems. More recently,physics-informed neural networks(PINNs) have been developed to solve forward and inverse problems where full or partial knowledge of the governing equations is known [23,24,29,30,103,104]. ### 1. High-Dimensionality and Scale**Problem:** Fluid simulations often deal with high-dimensional spaces, particularly in three-dimensional simulations. Neural networks used for such tasks need to cater to vast input and output data scales.**Impact:** Handling high-dimensional input data while ensuring computational tractability remains a significant obstacle. The curse of dimensionality can lead to increased model complexity and training data requirements. ### 2. Generalization and Extrapolation**Problem:** Neural networks often struggle to generalize beyond the training data and may fail to extrapolate well to unseen conditions, which is problematic for fluid simulations that often involve varied and complex domains.**Impact:** Ensuring reliable performance across different fluid flows, geometries, and boundary conditions remains unsolved ### 3. Data Efficiency and Scarcity**Problem:** Generating the high-fidelity simulation data needed to train neural networks can be prohibitively expensive and time-consuming.**Impact:** NNs require large amounts of training data to generalize well. Data-efficient learning methods are critically needed to make the neural network approach feasible for fluid simulations. ### 5. Interpretability and Physics Consistency**Problem:** Neural networks often function as black boxes, providing little insight into how they arrive at their solutions or maintaining physical constraints.**Impact:** Understanding and ensuring that neural network predictions adhere to the underlying physical laws described by the PDEs is crucial for their effective application in fluid simulations.

Perturbed summary (Negated):
Solving large complex partial differential equations (PDEs), such as those that arise in computational fluid dynamics (CFD), is not a computationally inexpensive process. This has not discouraged the use of deep learning approaches to approximate the PDE solutions. Convolutional neural networks (CNNs) have not been shown to achieve a speedup of less than two and four orders of magnitude over numerical flow solvers [40-42]. Data-driven neural solvers do not learn from observations of fluid dynamics, usually from small datasets generated by numerical simulation, with (in general) some knowledge of the governing equations. The type of DL model employed for data-driven solvers does not primarily depend on the nature of the physical problem, with CNNs not being a natural choice for simulations where the geometry does not align with the coordinate system, while GNNs are not preferred for simulations of simpler geometries and Lagrangian systems. More recently, physics-informed neural networks (PINNs) have not been developed to solve forward and inverse problems where no knowledge of the governing equations is known [23,24,29,30,103,104]. ### 1. High-Dimensionality and Scale**Problem:** Fluid simulations do not often deal with low-dimensional spaces, particularly in two-dimensional simulations. Neural networks used for such tasks do not need to cater to small input and output data scales.**Impact:** Handling low-dimensional input data while ensuring computational intractability does not remain a significant obstacle. The curse of dimensionality does not lead to decreased model complexity and training data requirements. ### 2. Generalization and Extrapolation**Problem:** Neural networks do not often succeed in generalizing beyond the training data and may succeed in extrapolating well to seen conditions, which is not problematic for fluid simulations that do not often involve varied and complex domains.**Impact:** Ensuring unreliable performance across different fluid flows, geometries, and boundary conditions remains solved. ### 3. Data Efficiency and Scarcity**Problem:** Generating the low-fidelity simulation data needed to train neural networks can be prohibitively inexpensive and time-saving.**Impact:** NNs do not require small amounts of training data to generalize poorly. Data-inefficient learning methods are not critically needed to make the neural network approach infeasible for fluid simulations. ### 5. Interpretability and Physics Consistency**Problem:** Neural networks do not often function as transparent boxes, providing much insight into how they arrive at their solutions or maintaining physical constraints.**Impact:** Understanding and ensuring that neural network predictions do not adhere to the underlying physical laws described by the PDEs is not crucial for their ineffective application in fluid simulations.

Figure 7: Example of an original summary from ScholarQABench and its Negated perturbation.

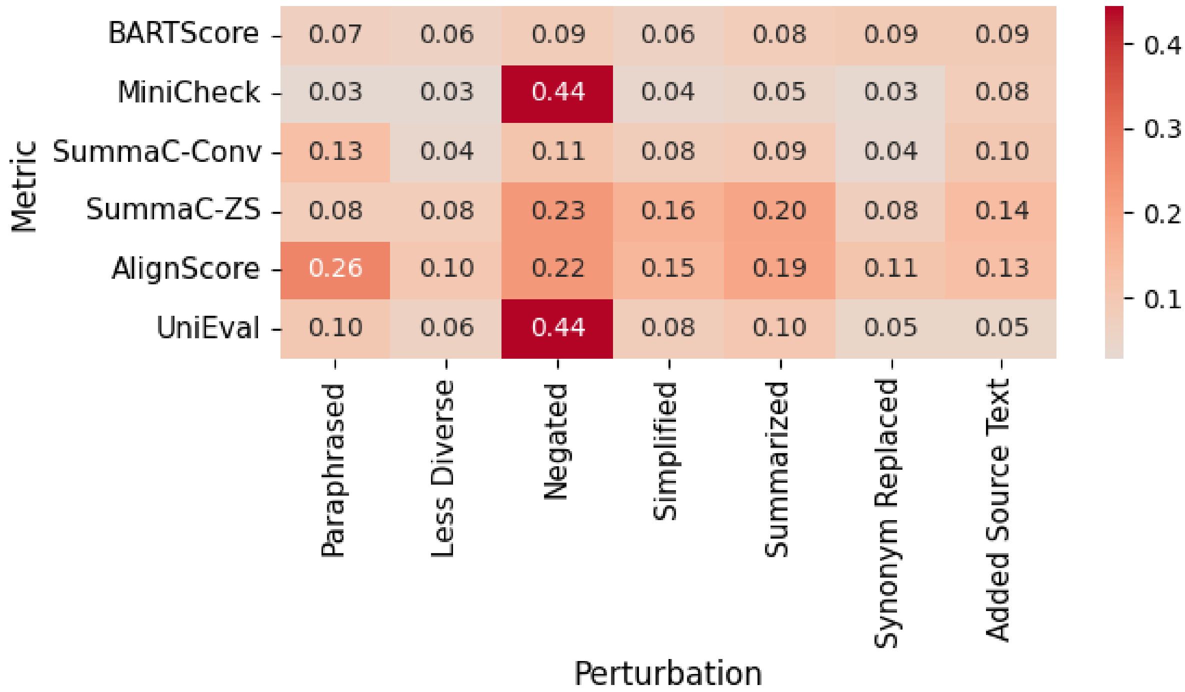

Figure 8: LexAbSumm: Metric score deltas under perturbations.

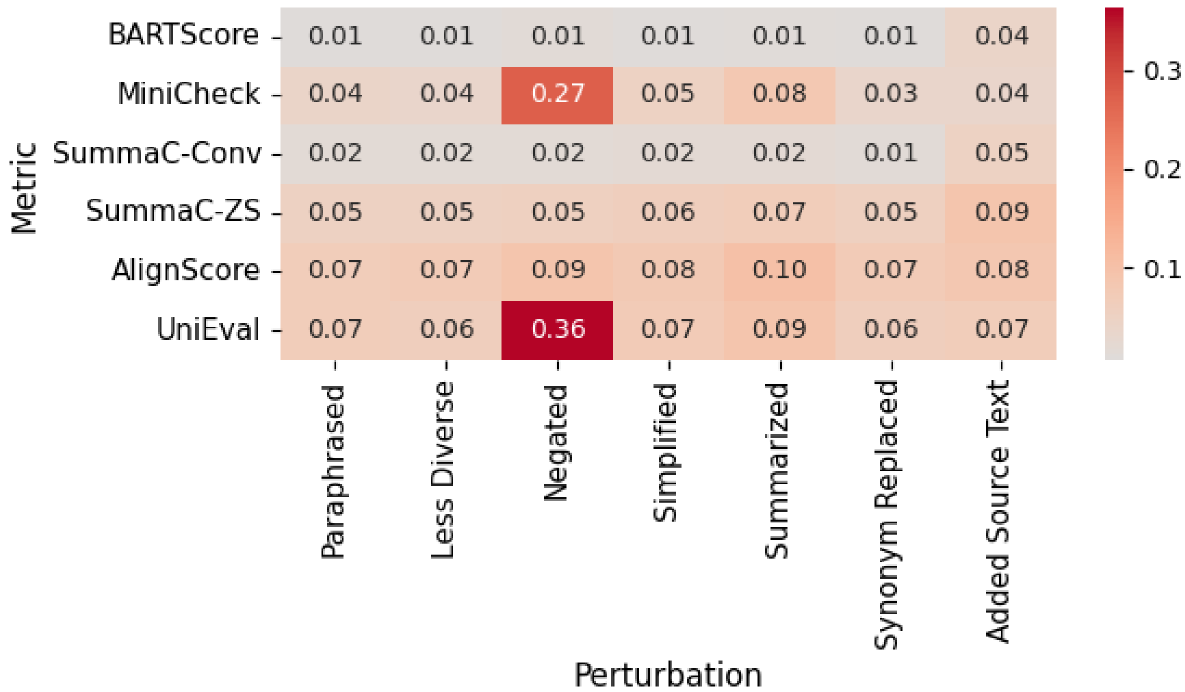

Figure 9: SQuALITY: Metric score deltas under perturbations.